

Braid Presentations for Concurrent Programs

Zachary Stone

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Abstract

Ziemiański’s cube chain category $\text{Ch}(K)$ is a combinatorial model for the space of directed paths on a precubical set K . Paliga and Ziemiański built a braid invariant for precubical sets by observing a relation between $\pi_1|\text{Ch}(Z)|$ and braids, where Z is the terminal precubical set. However, that invariant trivializes for Euclidean K because $|\text{Ch}(I^n)|$ is contractible [PZ21, Prop. 6.1]. This paper constructs a localization of $\text{Ch}(K)$ at morphisms that preserve the order of coordinates, called $\text{Br}(K)$. And we will show that this structure is the source of the braid structure of $\pi_1|\text{Ch}(Z)|$. But, there is no nerve or geometric realization in this technique. Instead we build a 2-polygraph giving an artin braid-like presentation of $\text{Br}(K)$. The $\{0, 1, 2\}$ -cells of the polygraph are the dimension $\{0, 1, 2\}$ chains of K . Intuitively, 3-cells of K contribute braid relations, and pairs of non-intersecting 2-cells of K contribute commuting relations. This is not just a metaphor: at $K = Z$, we get the artin relations and generators exactly. So $\text{Br}(Z)$ is equivalent to the positive braid category. The presentation is natural in K , providing a generalization of the existing work. As an application, we establish a link to the hyperplane arrangement literature. The way braids are constructed via Salvetti’s theorem for the braid arrangement exactly matches the polygraph we build here for the cube. All this work is formalized in Lean [Sto26].

1 Introduction

Ziemiański’s cube chain category $\text{Ch}(K)$ models the executions of a bipointed precubical set K [Zie20, Thm. 7.6], and Paliga–Ziemiański’s homotopy equivalence between $|\text{Ch}(Z)|$ and configuration spaces of points in the plane makes it a braid invariant of any K , by naturality. It trivializes on Euclidean complexes [PZ21, Prop. 6.1] for a one-line reason: it factors through the contractible $|\text{Ch}(I^n)|$.

The braiding is already present one level down, in $\text{Ch}(K)$ itself, with no nerve. An object of $\text{Ch}(K)$ decomposes an execution into blocks of simultaneous events; a morphism is a *refinement*, a relabelling of events that coarsens the blocking. Some refinements only glue adjacent blocks and leave every event where it was; the rest genuinely reorder events across a block boundary. Call the first kind *order preserving*, write W for the class they form, and set $\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}]$. Nothing is quotiented: inverting W leaves a category whose arrows remember which events crossed which, and how often. It is a braiding, not a permutation — two independent events that swap twice are not the identity.

We present $\text{Br}(K)$ by generators and relations, for every K , naturally in K , and the generators and relations are not invented: they are the low-dimensional chains of K . A chain of degree 0 is a *run*, an execution with a total order on its events, and a chain e of any degree spans two runs, the run $\underline{e}$ it is glued from by an order-preserving refinement and the run $\bar{e}$ that reverses every block of e . So

chains are already arrow-shaped.

Let the **paper polygraph** $\mathcal{P}(K)$ (Definition 4.12) have 0-cells the runs of K , 1-cells $X \rightarrow Y$ the chains e of degree 1 with $\underline{e} = X$ and $\bar{e} = Y$, and 2-cells $X \rightarrow Y$ the chains of degree 2 likewise, a 2-cell e being attached along the two maximal chains of the poset of runs over e (Proposition 4.11).

Theorem 1.1 (Theorem 5.5, Theorem 6.1, Theorem 7.6). *For every bipointed precubical set K , with no hypothesis on K , the polygraph $\mathcal{P}(K)$ presents $\text{Br}(K)$. The assignment $K \mapsto \mathcal{P}(K)$ is a functor $\square_* \rightarrow \mathbf{Poly}$ and the presentation is natural in K , on the nose. At the terminal object, $\mathcal{P}(Z)$ is Artin’s polygraph cell for cell and word for word, so $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$.*

No braid theory goes in. Artin’s dichotomy is the dichotomy between the two kinds of degree-two chain: one block of dimension 3 spans a hexagon of runs and gives $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$, two blocks of dimension 2 span a square and give $\sigma_i \sigma_k = \sigma_k \sigma_i$ (Proposition 4.11). So 3-cells of K contribute braid relations and disjoint pairs of 2-cells contribute commutations. The nearest existing statement is Gaussent–Guiraud–Malbos’s coherent presentation of an Artin monoid [GGM15], from Garside theory; here the same 2-cells come from the geometry of K , and the Artin case is the value at a point.

Formalisation. Everything below is formalised in Lean 4 on mathlib [Sto26], sorry-free; $\text{Br}(K)$, W and $\mathcal{P}(K)$ are `ChainCat.Br`, `ChainCat.W` and `Paper.poly`, and Theorem 5.5, Theorem 6.1, Theorem 7.6 are `Paper.paperPresents`, `Paper.paperSquare` and `Paper.paperArtinIso`. The file `CubeChains/Paper.lean` is this paper read back into the tree, each statement below appearing there as an `example` discharged by one citation.

Notation. **Cube** is the box category (objects the cubes I^n , morphisms the face embeddings), $\square = \widehat{\mathbf{Cube}}$ the precubical sets, and $\square_*$ the bipointed ones, with Z final and I^n bipointed by $(0, \dots, 0)$ and $(1, \dots, 1)$. We set $[n] := \{1, \dots, n\}$, write I^n both for the cube and for the presheaf it represents, and identify $K_n = \text{Hom}_{\square}(I^n, K)$ by Yoneda. Named ambient categories are bold (**Cube**, **Braid**⁺, **Poly**), constructions applied to an object upright ($\text{Ch}(K)$, $\text{Sq}(d)$), posets and polygraphs calligraphic ($\mathcal{R}(d)$, $\mathcal{P}(K)$). Composition is classical, so gf means “first f , then g ”, while a word in a polygraph is a path read left to right, so $\sigma_i \sigma_k$ is the arrow $\sigma_k \circ \sigma_i$.

Two ^{op}s are bookkeeping. Following [PZ21; Zie20] a morphism of $\text{Ch}(K)$ runs from a *finer* chain to a coarser one, against the direction in which an execution is built up, so we localize the opposite category and a word of cells then composes in the reading direction; and the same convention makes the endomorphism monoid of a loop polygraph the opposite of the monoid its words spell, whence **Braid**^{+, op} rather than **Braid**⁺, which word reversal identifies with it.

2 Background

Cube chains [PZ21; Zie20]. The **wedge** $P \vee Q$ of $P, Q \in \square_*$ is the pushout in $\square$ gluing fin_P to init_Q , bipointed by init_P and fin_Q ; $(\square_*, \vee, I^0)$ is monoidal. A **shape** is a list $a = [a_1, \dots, a_r]$ of positive naturals, with **serial wedge** $\bigvee a := I^{a_1} \vee \dots \vee I^{a_r}$, **beads** the I^{a_i} , and $|a| := \sum_i a_i$, $\deg a := \sum_i (a_i - 1) = |a| - r$; write $\square^\vee \subseteq \square_*$ for the full subcategory of serial wedges. Then $\text{Ch}(K)$ is the full subcategory of $\square_*/K$ spanned by the serial wedges: an object is a bipointed $\bigvee a \rightarrow K$, a morphism (a **refinement**) is a map of wedges commuting over K , running from a finer chain to a coarser one. We write $\deg d$, $|d|$ for the degree and event count of d ’s shape and $\text{Ch}(u) : \text{Ch}(K) \rightarrow \text{Ch}(L)$ for the pushforward along $u : K \rightarrow L$, which postcomposes the classifying

map and changes no shape. A chain of degree 0 is a **run**; $\text{Run}(K) \subseteq \text{Ch}(K)$ is the full subcategory of runs, and it is discrete, since a refinement losing no bead is an identity.

We use three facts from [PZ21, §2]. Serial wedges are non-self-linked and admit an altitude function $\text{alt} : K_0 \rightarrow \mathbb{Z}$ with $\text{alt}(\text{fin } e) = \text{alt}(\text{init } e) + 1$ on edges, the final vertex of $\bigvee a$ having altitude $|a|$; a bipointed map between connected such objects preserves altitude [PZ21, Prop. 2.7]; and Ch is lax monoidal from $(\square_*, \vee)$ to $(\mathbf{Cat}, \times)$ by concatenation, with $\text{Ch}(P) \times \text{Ch}(Q) \rightarrow \text{Ch}(P \vee Q)$ an isomorphism when $P \vee Q$ is non-self-linked and admits altitude [PZ21, Lem. 2.11].

The events of a wedge. Let $\text{Coord} : \mathbf{Cube} \rightarrow \mathbf{Set}$ send I^n to $[n]$ and a face embedding g to the monotone injection g_* naming the directions g leaves free, and let $E(\text{Coord})$ be its left Kan extension along Yoneda. It is cocontinuous and $\text{Coord}(I^0) = \emptyset$, so the pushout defining the wedge becomes a coproduct:

$$\text{Ev}(a) := E(\text{Coord})\left(\bigvee a\right) \cong \coprod_i [a_i],$$

the **events** of a . Write $\text{blk}(p)$ for the bead index of an event, $f_* := E(\text{Coord})(f)$, and $\text{str}_a : \text{Ev}(a) \cong [|a|]$ for the order isomorphism from the lexicographic order (bead first, then direction); $\text{str}_a(p)$ is the **strand** of p , its position in the canonical execution.

Proposition 2.1. *Let $f : \bigvee a \rightarrow \bigvee b$ in $\square_*$. Then f is determined by f_* , which is a bijection, and, writing $\text{blk}(f)$ for the induced map on bead indices,*

$$f_*(i, d) = (\text{blk}(f)(i), (f_i)_*(d)), \quad \text{blk} \circ f_* = \text{blk}(f) \circ \text{blk}, \quad (1)$$

with $\text{blk}(f)$ monotone and each $f_i : I^{a_i} \rightarrow I^{b_{\text{blk}(f)(i)}}$ a face embedding.

Proof. By the pushout, f is determined by its restrictions to the beads, and each lands in one bead of the target since $a_i > 0$: that gives $\text{blk}(f)$, (1) and the last clause, and a face embedding is determined by the directions it leaves free. Injectivity of f_* and monotonicity of $\text{blk}(f)$ follow by induction on the length of b : by [PZ21, Lem. 2.11] a map $\bigvee a \rightarrow I^m \vee \bigvee b'$ splits as $f_1 \vee f_2$ over a splitting $a = a_l * a_r$, so $f_* = (f_1)_* \sqcup (f_2)_*$ and $\text{blk}(f) = \text{blk}(f_1) \sqcup (1 + \text{blk}(f_2))$ with $\text{blk}(f_1)$ constant at 1, and injectivity of $(f_1)_* : \text{Ev}(a_l) \rightarrow [m]$ is a second induction, the images at each glued vertex separated by the coordinates already 1 there. Surjectivity is a count: f preserves altitude [PZ21, Prop. 2.7], so $|a| = |b|$. $\square$

Polygraphs, localization, and Newman's lemma. A **2-polygraph** $\mathcal{P}$ is a set V of 0-cells, families $\mathcal{P}_1(x, y), \mathcal{P}_2(x, y)$ of 1- and 2-cells, and two *words* $\text{src}, \text{tgt} : \mathcal{P}_2(x, y) \rightarrow \{\text{paths } x \rightarrow y\}$ in the free category on the 1-cells, read left to right; morphisms are maps in each dimension commuting with src, tgt on the nose, and **Poly** is the resulting category. The category $\langle \mathcal{P} \rangle$ **presented by** $\mathcal{P}$ is the free category on the 1-cells modulo the congruence generated by $\text{src}(\alpha) = \text{tgt}(\alpha)$, with quotient functor q ; $\mathcal{P}$ **presents** C when given an equivalence $\langle \mathcal{P} \rangle \rightarrow C$. We use its universal property [Ara+23, Ch. 2] and that of the localization [GZ67, Ch. I] in the following form.

Proposition 2.2. *A functor $\langle \mathcal{P} \rangle \rightarrow C$ is the same as a map on 0-cells together with an arrow of C per 1-cell sending the two words of each 2-cell to the same arrow; two such functors agreeing on 0- and 1-cells are equal; and a morphism of polygraphs bijective in every dimension is an isomorphism. For a class W in C there is $Q : C \rightarrow C[W^{-1}]$ inverting W , through which every functor inverting W factors uniquely, and two functors out of $C[W^{-1}]$ agreeing after Q are equal.*

For $m : a \rightarrow e$ in W and $f : b \rightarrow e$ arbitrary we write $\text{frac}(m, f) := Q(f^{\text{op}}) \circ Q(m^{\text{op}})^{-1} : Q(a) \rightarrow Q(b)$ in $C^{\text{op}}[(W^{\text{op}})^{-1}]$, “back along the order-preserving leg, forward along the other”. Then $\text{frac}(m, m) = \text{id}$ and two such glue: for $l : e_0 \rightarrow e$, $m_0 : b \rightarrow e_0$ in W and $f_0 : c \rightarrow e_0$,

$$\text{frac}(m_0, f_0) \circ \text{frac}(m, l \circ m_0) = \text{frac}(m, l \circ f_0). \quad (2)$$

Write $\text{Hasse}(P)$ for the Hasse quiver of a poset P , and call a map sending each element to an object of C and each cover to an arrow a **reading**; it extends to paths.

Theorem 2.3 (coherent Newman; [Hue80], [Ara+23, Ch. 4]). *Let P be finite and φ a reading of P in C such that for all covers $x \leq y_1$, $x \leq y_2$ with $y_1 \neq y_2$ and every common upper bound z of y_1, y_2 , there are $j \leq z$ and paths $p_i : y_i \rightarrow j$ with $\varphi(x \leq y_1)\varphi(p_1) = \varphi(x \leq y_2)\varphi(p_2)$. Then φ sends parallel paths to equal arrows, so it extends uniquely to a functor $P \rightarrow C$, and a functor out of P is pinned by its values on covers.*

Lemma 2.4. *If a finite poset P is graded by r with $r(y) = r(x) + 1$ on covers and every element has exactly two Hasse neighbours, then a directed path is determined by its first edge; and if P has a least and a greatest element then P is a cycle, its two maximal chains have a common length ℓ , and $|P| = 2\ell$. (Undirected, the Hasse graph is 2-regular hence a union of cycles; r increases along a path, so it leaves each vertex by the edge it did not arrive on; both edges at $\perp$ point up and both at $\top$ point down, so the cycle through $\perp$ meets $\top$ and splits into two arcs of equal length, and it is all of P .)*

3 Squares, runs, and the braiding

3.1 Squares, triangles, and biclosed sets

Definition 3.1. Let $\text{Sq}, \text{Tri} : \mathbf{Cube} \rightarrow \mathbf{Set}$ send I^n to the monotone injections $[2] \hookrightarrow [n]$, respectively $[3] \hookrightarrow [n]$. Both vanish at I^0 , so their left Kan extensions along Yoneda give, as for Ev ,

$$\text{Sq}(a) \cong \coprod_i \binom{[a_i]}{2}, \quad \text{Tri}(a) \cong \coprod_i \binom{[a_i]}{3}, \quad \text{so} \quad |\text{Sq}(a)| = \sum_i \binom{a_i}{2}.$$

Concretely a **square** s is a pair of events of one bead, written $\text{lo}(s), \text{hi}(s)$ with $\text{str}(\text{lo } s) < \text{str}(\text{hi } s)$, and a **triangle** t is a triple of events of one bead, with edges $t_{01}, t_{02}, t_{12} \in \text{Sq}(a)$. We write $\text{Sq}(d)$ for the squares of a chain’s shape and $q_* : \text{Sq}(o) \rightarrow \text{Sq}(d)$ for the map induced by a refinement, injective by Proposition 2.1. A set $R \subseteq \text{Sq}(a)$ is **biclosed** when for every triangle t ,

$$t_{01}, t_{12} \in R \implies t_{02} \in R, \quad \text{and} \quad t_{02} \in R \implies t_{01} \in R \text{ or } t_{12} \in R;$$

equivalently, both R and R^c are closed under the first rule.

Lemma 3.2. *For $q : \bigvee a \rightarrow \bigvee b$, a square s of $\bigvee b$ lies in $\text{im } q_*$ iff $q_*^{-1}(\text{lo } s)$ and $q_*^{-1}(\text{hi } s)$ share a bead of $\bigvee a$; and for events u, v of $\bigvee b$ in distinct beads, $\text{blk}(q_*^{-1}u) < \text{blk}(q_*^{-1}v) \iff \text{blk}(u) < \text{blk}(v)$.*

Proof. If $s = q_*t$ the events pull back to t ’s, which share a bead; conversely if they share a bead i then, $(q_i)_*$ being a monotone injection, they are the events of a square of that bead. The second claim is monotonicity of $\text{blk}(q)$ (Proposition 2.1) in both directions, the beads being distinct. $\square$

3.2 Runs of a wedge

A run over the wedge $\vee a$ is a chain $x : \vee[1^N] \rightarrow \vee a$ of degree 0, where $N = |a|$; by Proposition 2.1 it is determined by the bijection $x_* : [N] = \text{Ev}([1^N]) \rightarrow \text{Ev}(a)$. Write $\text{step}_x := x_*^{-1}$ for its **firing order** and

$$\text{rev}(x) := \{ s \in \text{Sq}(a) : \text{step}_x(\text{hi } s) < \text{step}_x(\text{lo } s) \}$$

for the squares it **reverses**.

Lemma 3.3. *A run fires the beads in blocks: $\text{blk}(p) < \text{blk}(q)$ implies $\text{step}_x(p) < \text{step}_x(q)$. So a run is exactly a choice of linear order on $[a_i]$ for each bead, and blk is a monotone surjection from the firing order onto the beads, with interval fibres.*

Proof. The beads of $[1^N]$ are singletons, so blk there is the identity and $\text{blk}(x) = \text{blk} \circ x_*$, monotone by Proposition 2.1. A monotone surjection from a chain has interval fibres, here the beads; the only freedom left is the order inside each. $\square$

Theorem 3.4 (a run is a biclosed set). *$x \mapsto \text{rev}(x)$ is a bijection from the runs of $\vee a$ onto the biclosed subsets of $\text{Sq}(a)$. Reversing every bead's order complements rev ; in particular biclosed sets are closed under complement.*

Proof. A triangle lies in one bead, so by Lemma 3.3 the statement is bead-local; fix a bead $[m]$ with its cube order $<$. A linear order $\prec$ on $[m]$ has inversion set $\text{rev}(\prec) = \{(p, q) : p < q, q \prec p\}$, from which it is recovered by: $p \prec q$ iff $(p, q) \notin R$ or $(q, p) \in R$. So rev is injective, and its image is the set of R for which that recipe is transitive — totality being automatic. Transitivity can only fail on a triple $p < q < r$, where the cases to check are exactly the two biclosedness clauses at that triangle. Complementation reverses each $\prec$. $\square$

For one bead of dimension m this is the classical description of inversion sets in S_m , and inclusion of reversed sets is the right weak order [BB05, §3.1]; the order theory below is that statement, bead by bead.

Lemma 3.5. *Order $\text{Sq}(a)$ by $\text{str}(\text{lo } -)$, then by $\text{str}(\text{hi } -)$; call this the **standard order**. Its lower sets and its upper sets are all biclosed.*

Proof. On a triangle with events $p < q < r$ the edges are $t_{01} = (p, q)$, $t_{02} = (p, r)$, $t_{12} = (q, r)$, in that standard order. For a lower set R : $t_{12} \in R$ forces $t_{02} \in R$ and $t_{02} \in R$ forces $t_{01} \in R$, which are the two clauses. For an upper set: $t_{01} \in R$ forces $t_{02} \in R$ and $t_{02} \in R$ forces $t_{12} \in R$. $\square$

3.3 Crossings, order-preserving refinements, and $\text{Br}(K)$

Definition 3.6. A map of serial wedges $\varphi : \vee a \rightarrow \vee b$ **crosses**

$$\text{Cr}(\varphi) := \{ s \in \text{Sq}(b) : \text{blk}(\varphi_*^{-1}(\text{hi } s)) < \text{blk}(\varphi_*^{-1}(\text{lo } s)) \},$$

the squares of the target whose upper event it brings from a strictly earlier bead than their lower. For a refinement we write $\text{Cr}(f)$ for the crossings of its wedge map.

Lemma 3.7 (the cocycle). *$\text{Cr}(\psi\varphi) = \psi_*(\text{Cr } \varphi) \sqcup \text{Cr}(\psi)$, disjointly, for $\varphi : \vee a \rightarrow \vee b$ and $\psi : \vee b \rightarrow \vee c$; and $\text{Cr}(\text{id}) = \emptyset$.*

Proof. Take $s \in \text{Sq}(c)$. If $s = \psi_*(t)$ then $s \notin \text{Cr}(\psi)$, since t 's events share a bead (Lemma 3.2) — which also gives disjointness — and $(\psi\varphi)_*^{-1}$ agrees with φ_*^{-1} on t 's events, so $s \in \text{Cr}(\psi\varphi)$ iff $t \in \text{Cr}(\varphi)$. If $s \notin \text{im } \psi_*$ then $s \notin \psi_*(\text{Cr } \varphi)$ and, the events $\psi_*^{-1}(\text{lo } s)$, $\psi_*^{-1}(\text{hi } s)$ lying in distinct beads, the second part of Lemma 3.2 applied to φ gives $s \in \text{Cr}(\psi\varphi)$ iff $s \in \text{Cr}(\psi)$. $\square$

Definition 3.8. A refinement f is **order preserving** when $\text{Cr}(f) = \emptyset$; write $W = W_K$ for the class of these, and

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}], \quad [c] := Q(c).$$

Since W is a condition on the wedge map alone, $\text{Ch}(u)$ carries W_K into W_L and descends to $\text{Br}(u) : \text{Br}(K) \rightarrow \text{Br}(L)$, strictly functorially in u .

By Proposition 3.9 below, $f \in W$ means f only glues adjacent beads; the elementary ones are the comparisons $\vee[\dots, p, q, \dots] \rightarrow \vee[\dots, p + q, \dots]$, and W is the multiplicative class they generate.

Proposition 3.9. *For a refinement $f : c \rightarrow d$ the following agree: $f \in W$; f_* is monotone for the strand orders; $\text{str}_d \circ f_* = \text{str}_c$, i.e. every event keeps its position. Consequently W contains the identities, $fg \in W$ iff $f \in W$ and $g \in W$, and two order-preserving refinements with the same source and target are equal.*

Proof. A monotone bijection of finite chains of equal size is the unique isomorphism, which gives the second and third conditions from each other. For the first two, take $u, v \in \text{Ev}(c)$ with $\text{str}(u) < \text{str}(v)$. If $\text{blk}(u) = \text{blk}(v)$ then f_*u, f_*v lie in one bead in the same order, $(f_i)_*$ being a monotone injection. If $\text{blk}(u) < \text{blk}(v)$ then $\text{blk}(f_*u) \leq \text{blk}(f_*v)$, so $\text{str}(f_*u) > \text{str}(f_*v)$ can happen only when both land in one bead, and then $s := (f_*v, f_*u)$ is a square with $\text{blk}(f_*^{-1}\text{hi } s) < \text{blk}(f_*^{-1}\text{lo } s)$, that is $s \in \text{Cr}(f)$; conversely every crossing arises so. Closure and two-out-of-three are Lemma 3.7 with ψ_* injective, and the last claim holds because the third condition pins $f_* = \text{str}_d^{-1} \text{str}_c$, which pins f (Proposition 2.1). $\square$

4 The paper polygraph

Fix K , and let d be a chain of K of shape a , with $N = |d|$.

4.1 Runs over a chain

Definition 4.1. A **run over** d is a run X of K with a refinement $X \rightarrow d$; write $\mathcal{R}(d)$ for the set of them, $\text{run}(x)$ and $\text{ref}(x)$ for the two components, and $q_\#x := q \circ \text{ref}(x)$ for the **push** along a refinement $q : o \rightarrow d$.

Lemma 4.2. *Composing with d 's classifying map identifies $\mathcal{R}(d)$ with the runs of $\vee a$, and under that identification $\text{rev}(x) = \text{Cr}(\text{ref}(x))$. So, ordering $\mathcal{R}(d)$ by $x \leq y : \iff \text{rev}(x) \subseteq \text{rev}(y)$, the map rev is an isomorphism of posets onto the biclosed subsets of $\text{Sq}(d)$, with least element $\perp_d$ ($\text{rev} = \emptyset$), greatest element $\top_d$ ($\text{rev} = \text{Sq}(d)$), and complementation as an anti-automorphism. Write $\underline{d}, \bar{d} \in \text{Run}(K)$ for the runs underlying $\perp_d$ and $\top_d$.*

Proof. A refinement $X \rightarrow d$ over K is a wedge map $\vee[1^N] \rightarrow \vee a$ whose composite with d 's classifying map is X 's, so the latter is forced and the data is just the wedge map. Every bead of the source is then an edge, so blk on the source is str and $\text{blk} \circ \text{ref}(x)_*^{-1} = \text{step}_x$, which makes Definition 3.6 and the display above the same condition. The rest is Theorem 3.4. $\square$

Proposition 4.3 (the push formula). $\text{rev}(q_{\#}x) = q_*(\text{rev } x) \sqcup \text{Cr}(q)$. Hence $q_*^{-1}\text{rev}(q_{\#}x) = \text{rev}(x)$, so $q_{\#}$ is an order embedding; and $q_{\#}\perp_o = \perp_d$ iff $q \in W$. Moreover, for $x \in \mathcal{R}(d)$ the conditions $\text{ref}(x) \in W$, $\text{rev}(x) = \emptyset$ and $x = \perp_d$ agree, so $\underline{X} = X$ for a run X , and $\underline{d} = \underline{c}$ whenever $c \rightarrow d$ lies in W .

Proof. The formula is Lemma 3.7 read through Lemma 4.2. Since q_* is injective with image missing $\text{Cr}(q)$ (Lemma 3.2), it follows that $q_*^{-1}\text{rev}(q_{\#}x) = \text{rev}(x)$ and that $q_{\#}x \leq q_{\#}y$ iff $x \leq y$; and $\text{rev}(q_{\#}\perp_o) = \text{Cr}(q)$. The last sentence is Lemma 4.2 and the formula applied to $c \rightarrow d$. $\square$

4.2 Steps, covers, and cells

Lemma 4.4. Call $s \in \text{Sq}(d)$ a **step** of $x \in \mathcal{R}(d)$ when $\text{rev}(x) \triangle \{s\}$ is biclosed, write $\text{St}(x)$ for the set of them and x^s for the **flip**, the run reversing $\text{rev}(x) \triangle \{s\}$. Then s is a step iff its two events are consecutive in the firing order of x , and $|\text{St}(x)| = \deg d$.

Proof. A bead's events are consecutive in the firing order (Lemma 3.3), so “consecutive in the bead” and “consecutive overall” agree. If the two events are consecutive, toggling s swaps an adjacent pair of a linear order, so the result is biclosed by Theorem 3.4; if some event w of the bead fires strictly between them, toggling swaps them but not their order against w , so w would lie both before and after the pair. Thus the steps are the covers of the firing order lying inside a bead. That order is a chain on N elements and blk is a monotone surjection onto the r beads (Lemma 3.3); such a map collapses exactly $N - r$ covers, so there are $N - r = \deg d$ of them. $\square$

Lemma 4.5 (the cover lemma). If $x < z$ then some $s \in \text{St}(x) \setminus \text{rev}(x)$ lies in $\text{rev}(z)$. Consequently $x < y$ iff $\text{rev}(y) = \text{rev}(x) \cup \{s\}$ for such an s , in which case $y = x^s$; so $\mathcal{R}(d)$ is graded by $|\text{rev}|$, from 0 at $\perp_d$ to $|\text{Sq}(d)|$ at $\top_d$.

Proof. Write $\prec_x, \prec_z$ for the firing orders. If z agreed with x on every x -covering pair then $\prec_x \subseteq \prec_z$ by transitivity, so $z = x$. So some x -cover $p \prec_x q$ has $q \prec_z p$; its events share a bead, since otherwise $\text{blk}(p) < \text{blk}(q)$ and every run fires p first (Lemma 3.3). So the pair is a step s of x (Lemma 4.4); if $\text{lo } s = p$ then $s \in \text{rev}(z) \setminus \text{rev}(x)$ as wanted, and if $\text{lo } s = q$ then $s \in \text{rev}(x) \subseteq \text{rev}(z)$ says $p \prec_z q$, a contradiction. For the covering criterion, $x \leq x^s \leq y$ forces $y = x^s$ in one direction, and in the other two biclosed sets differing by one element are a covering pair under Lemma 4.2. $\square$

Proposition 4.6 (a factorisation is a set of steps). Sending a factorisation $\text{run}(x) \rightarrow m \rightarrow d$ of $\text{ref}(x)$ to the set of steps of x whose events share a bead of m is a bijection onto the subsets of $\text{St}(x)$, with $\deg m = |S|$. Write m_S for the middle and ι_S, π_S for the legs. If moreover $S \cap \text{rev}(x) = \emptyset$ — we then say S **ris**es from x — then $\iota_S \in W$, so $\underline{m}_S = \text{run}(x)$.

Proof. A middle m makes the N events into its beads, so it cuts the firing order of x into intervals (Lemma 3.3 for x over m), and π requires each interval to sit inside one bead of d . Conversely such an interval partition arises from the chain whose beads are its blocks in order; and the factorisation is pinned by that shape, because a wedge map is determined by its action on events and injective there (Proposition 2.1), hence monic, while the first leg is forced by which events fall in which block. An interval partition of a chain is the set of covers it does not cut, and those lie inside a bead, so they are steps (Lemma 4.4); and $\deg m = N - \#\text{blocks} = \#\text{uncut covers} = |S|$.

If S rises, then by Lemma 4.2 $\text{Cr}(\iota_S)$ is the set of squares of m_S that x reverses, that is the pairs inside one block that x reverses. The covers inside a block all lie in S , so none is reversed, so the

firing order agrees with the strand order on consecutive pairs of the block and hence, by transitivity, on all of them. So $\text{Cr}(\iota_S) = \emptyset$. $\square$

Definition 4.7. An n -cell $X \rightarrow Y$ is a chain e with $\deg e = n$, $\underline{e} = X$ and $\bar{e} = Y$; write $\text{Cell}_n(X, Y)$. So the n -cells are just the chains of degree n , sorted by the two runs each spans.

Lemma 4.8 (the closure lemma). *Let S rise from x . Then $\text{rev}((\pi_S)_\# \top_{m_S}) = \text{rev}(x) \cup (\pi_S)_*(\text{Sq}(m_S))$, and any $z \geq x$ with $S \subseteq \text{rev}(z)$ has $(\pi_S)_*(\text{Sq}(m_S)) \subseteq \text{rev}(z)$. So $(\pi_S)_\# \top_{m_S}$ is the least element above x reversing all of S .*

Proof. The push formula gives $\text{rev}((\pi_S)_\# \top_{m_S}) = (\pi_S)_* \text{Sq}(m_S) \cup \text{Cr}(\pi_S)$, and the cocycle applied to $\pi_S \iota_S = \text{ref}(x)$ with $\text{Cr}(\iota_S) = \emptyset$ gives $\text{Cr}(\pi_S) = \text{rev}(x)$. For the second claim, list a block as $p_1 < \dots < p_k$ in strand order — which is its firing order, as in the previous proof. Each (p_i, p_{i+1}) lies in S , so z reverses it; since $\text{rev}(z)$ is biclosed, the first clause at the triangles (p_i, p_{i+1}, p_j) gives by induction that z reverses every pair of the block. $\square$

Corollary 4.9 (covers are 1-cells). *For a cover $x \leq y$ with step s , the chain $\gamma(x \leq y) := m_{\{s\}}$ is a 1-cell $\text{run}(x) \rightarrow \text{run}(y)$, with $(\pi_{\{s\}})_\# \perp_{m_{\{s\}}} = x$ and $(\pi_{\{s\}})_\# \top_{m_{\{s\}}} = y$; conversely every 1-cell α is γ of the unique cover $\perp_\alpha \leq \top_\alpha$ of $\mathcal{R}(\alpha)$. So γ , with $x \mapsto \text{run}(x)$ on vertices, is a map of quivers $\text{Hasse}(\mathcal{R}(d)) \rightarrow (\text{Run}(K), \text{Cell}_1)$, and it is natural: $\gamma(q_\# e) = \gamma(e)$ for every refinement $q : o \rightarrow d$.*

Proof. $m_{\{s\}}$ has degree 1 and $\underline{m_{\{s\}}} = \text{run}(x)$ by Proposition 4.6; by Lemma 4.8 its top pushes to the least element above x reversing s , namely $x^s = y$. Conversely a chain α of degree 1 has $|\text{Sq}(\alpha)| = 1$, so $\mathcal{R}(\alpha) = \{\perp_\alpha \leq \top_\alpha\}$ and the factorisation through that one step is α itself. For naturality, the step of $q_\# e$ is q_* of the step of e (Proposition 4.3), and q_* carries the firing order of x to that of $q_\# x$, so the two factorisations cut the same partition and have the same middle. $\square$

4.3 The attachment of a 2-cell, and the polygraph

Lemma 4.10. *If $\prec$ is a linear order on $\text{Sq}(d)$ whose lower sets are all biclosed, the runs x_t reversing its first t squares form a maximal chain $\perp_d = x_0 \leq \dots \leq x_{|\text{Sq}(d)|} = \top_d$, and the $(t+1)$ -st square is a step of x_t . By Lemma 3.5 the standard order and its dual both qualify.*

Proof. Each x_t exists by Lemma 4.2, consecutive ones differ by one element, and Lemma 4.5 applies. $\square$

Proposition 4.11 (the polygon, and the two species). *Let $\deg e = 2$. Then every $x \in \mathcal{R}(e)$ has exactly two steps, hence exactly two Hasse neighbours; $\mathcal{R}(e)$ is a polygon with $2|\text{Sq}(e)|$ elements; and its only two maximal chains from $\perp_e$ to $\top_e$ are the sweeps of the standard order and of its dual. Write $\text{src}(e)$ and $\text{tgt}(e)$ for the words of 1-cells they read off under γ ; they have length $|\text{Sq}(e)|$ and differ in their first letter. Moreover e has either one bead of dimension 3, with $|\text{Sq}(e)| = 3$ and $\mathcal{R}(e)$ a hexagon, or two beads of dimension 2, with $|\text{Sq}(e)| = 2$ and $\mathcal{R}(e)$ a square.*

Proof. $|\text{St}(x)| = 2$ by Lemma 4.4, and the two flips are the two neighbours (Lemma 4.5, up or down according to whether the step is reversed). The two sweeps of Lemma 4.10 leave $\perp_e$ by different covers, the least and greatest squares of the standard order being distinct, so Lemma 2.4 applies with the grading $|\text{rev}|$. Finally $\deg e = 2$ forces the bead dimensions above 1 to be $\{3\}$ or $\{2, 2\}$, and Definition 3.1 counts $\binom{3}{2} = 3$ and $2\binom{2}{2} = 2$. $\square$

Definition 4.12. The **paper polygraph** $\mathcal{P}(K)$ has 0-cells $\text{Run}(K)$, which is a set since a refinement between chains of degree 0 is an identity; 1-cells $\text{Cell}_1(X, Y)$; 2-cells $\text{Cell}_2(X, Y)$; and the words src, tgt of Proposition 4.11. Its data is nothing but the chains of K of degree 0, 1 and 2; the only computation is the attachment, and it has exactly two answers.

Proposition 4.13 (diverging covers close). *Read in $\langle \mathcal{P}(K) \rangle$, the quiver map γ satisfies the hypothesis of Theorem 2.3 on $\mathcal{R}(d)$, so it extends uniquely to a functor $\Gamma_d : \mathcal{R}(d) \rightarrow \langle \mathcal{P}(K) \rangle$, and $\Gamma_d \circ q_\# = \Gamma_o$ for every refinement $q : o \rightarrow d$.*

Proof. Let $x \leq y_1, x \leq y_2$ be distinct covers, with steps $s_1 \neq s_2$, and z a common upper bound. Then $S = \{s_1, s_2\}$ rises from x , so $\alpha := m_S$ is a chain of degree 2, that is a 2-cell (Proposition 4.6), and $j := (\pi_S)_\# \top_\alpha$ satisfies $j \leq z$ by Lemma 4.8. By Proposition 4.11 the two maximal chains of $\mathcal{R}(\alpha)$ are $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$ and leave $\perp_\alpha$ by its two covers. Pushing along π_S carries $\perp_\alpha$ to x , $\top_\alpha$ to j , covers to covers (Proposition 4.3, Lemma 4.5) and keeps each cover's 1-cell (Corollary 4.9); so the two chains push to two paths $x \rightarrow j$ leaving by y_1 and by y_2 and reading as $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$, equal in $\langle \mathcal{P}(K) \rangle$. For naturality, $q_\#$ is monotone, so $\Gamma_d \circ q_\#$ and Γ_o are functors out of $\mathcal{R}(o)$ agreeing on covers, hence equal by Theorem 2.3. $\square$

5 Theorem A: the polygraph presents $\text{Br}(K)$

Theorem A compares two universal properties: a functor out of $\langle \mathcal{P}(K) \rangle$ is a reading of the degree-one chains identifying the two words of every degree-two chain (Proposition 2.2), and a functor out of $\text{Br}(K)$ is a functor on $\text{Ch}(K)^{\text{op}}$ inverting W (Proposition 2.2). We build each from the other.

Definition 5.1. A cell $\alpha \in \text{Cell}_n(X, Y)$ carries two legs, the refinements of its extreme runs over itself: $\alpha_\perp := \text{ref}(\perp_\alpha) : X \rightarrow \alpha$, which is order preserving, and $\alpha_\top := \text{ref}(\top_\alpha) : Y \rightarrow \alpha$. Put $E(X) := [X]$ and $E(\alpha) := \text{frac}(\alpha_\perp, \alpha_\top) : [X] \rightarrow [Y]$.

Lemma 5.2 (telescoping, and soundness). *For $z \in \mathcal{R}(d)$ put $\text{at}(z) := \text{frac}(\text{ref}(\perp_d), \text{ref}(z))$. Then $\text{at}(\perp_d) = \text{id}$ and $\text{at}(y) = E(\gamma(x \leq y)) \circ \text{at}(x)$ for every cover, so every path from $\perp_d$ to z reads as $\text{at}(z)$ under $E \circ \gamma$. In particular the two words of a 2-cell e both read as $\text{at}(\top_e)$, so E extends to a functor $E : \langle \mathcal{P}(K) \rangle \rightarrow \text{Br}(K)$.*

Proof. $\text{at}(\perp_d) = \text{frac}(m, m) = \text{id}$. For a cover with step s , put $\beta := \gamma(x \leq y)$ and $q := \pi_{\{s\}}$; by Corollary 4.9, $\text{ref}(x) = q\beta_\perp$ and $\text{ref}(y) = q\beta_\top$, so (2) with $l = q$ gives $E(\beta) \circ \text{at}(x) = \text{frac}(\text{ref}(\perp_d), q\beta_\top) = \text{at}(y)$. Both words of a 2-cell are paths $\perp_e \rightarrow \top_e$ (Proposition 4.11); now apply Proposition 2.2. $\square$

Definition 5.3. Define $\Theta : \text{Ch}(K)^{\text{op}} \rightarrow \langle \mathcal{P}(K) \rangle$ by $\Theta(d) := \underline{d}$ and, for $u : c \rightarrow d$ in $\text{Ch}(K)$, $\Theta(u^{\text{op}}) := \Gamma_d(\perp_d \leq u_\# \perp_c)$, an arrow $\underline{d} \rightarrow \underline{c}$.

Lemma 5.4. Θ is a functor and inverts W^{op} ; indeed $\Theta(u^{\text{op}})$ is an identity for $u \in W$. So it descends uniquely to $\Phi : \text{Br}(K) \rightarrow \langle \mathcal{P}(K) \rangle$ with $\Phi Q = \Theta$.

Proof. For $v : b \rightarrow c$ and $u : c \rightarrow d$ we have $\perp_d \leq u_\# \perp_c \leq u_\# v_\# \perp_b = (uv)_\# \perp_b$, using that $u_\#$ is monotone (Proposition 4.3); since Γ_d is a functor, $\Theta((uv)^{\text{op}})$ is $\Gamma_d(u_\# \perp_c \leq u_\# v_\# \perp_b) \circ \Theta(u^{\text{op}})$, and the left factor is $\Gamma_c(\perp_c \leq v_\# \perp_b) = \Theta(v^{\text{op}})$ by naturality (Proposition 4.13). For $u \in W$, $u_\# \perp_c = \perp_d$ (Proposition 4.3). Now apply Proposition 2.2. $\square$

Theorem 5.5 (Theorem A). *For every K , with no hypothesis on K , the functor E is an equivalence $\langle \mathcal{P}(K) \rangle \simeq \text{Br}(K)$, with inverse Φ . So the runs of K and its chains of degree one and two present*

$\text{Br}(K)$. It is an equivalence and no more: $\text{Br}(K)$ is not skeletal, a chain and the run below it becoming isomorphic there without being equal, and the runs are a skeleton.

Proof. $\Phi E = \text{id}$. On a 0-cell, $\Phi(E(X)) = \Theta(X) = \underline{X} = X$ (Proposition 4.3). On a 1-cell $\alpha : X \rightarrow Y$, reading a cospan back gives $\Phi(\text{frac}(m, f)) = \Theta(f^{\text{op}})\Theta(m^{\text{op}})^{-1}$, and $\Theta(m^{\text{op}})$ is an identity for $m \in W$, so $\Phi(E(\alpha)) = \Theta(\alpha_{\top}^{\text{op}})$. Now $(\alpha_{\top})_{\#}\perp_Y$ is $\alpha_{\top}$ itself, that is $\top_{\alpha}$, and $\mathcal{R}(\alpha)$ is the two-element poset $\perp_{\alpha} \leq \top_{\alpha}$, so $\Theta(\alpha_{\top}^{\text{op}}) = \Gamma_{\alpha}(\perp_{\alpha} \leq \top_{\alpha}) = q(\gamma(\perp_{\alpha} \leq \top_{\alpha})) = q(\alpha)$ by Corollary 4.9. Conclude by Proposition 2.2.

$E\Phi \cong \text{id}$. By Lemma 5.2, $E(\Theta(u^{\text{op}})) = \text{at}(u_{\#}\perp_c) = \text{frac}(\text{ref}(\perp_d), u \circ \text{ref}(\perp_c))$, which unfolds to $Q(\text{ref}(\perp_c)^{\text{op}}) \circ Q(u^{\text{op}}) \circ Q(\text{ref}(\perp_d)^{\text{op}})^{-1}$. So $E\Theta \cong Q$, with component at c the isomorphism $[c] \cong [c]$ that $\text{ref}(\perp_c) \in W$ becomes in the localization; by uniqueness of descent (Proposition 2.2), $E\Phi \cong \text{id}_{\text{Br}(K)}$. $\square$

6 Theorem B: naturality in K

Theorem 6.1 (Theorem B). $K \mapsto \mathcal{P}(K)$, with $\mathcal{P}(f) := \text{Ch}(f)$ on cells in every dimension, is a functor $\square_* \rightarrow \mathbf{Poly}$, and for every $f : K \rightarrow L$ the presentation square commutes as an equality of functors:

$$E_L \circ \langle \mathcal{P}(f) \rangle = \text{Br}(f) \circ E_K : \langle \mathcal{P}(K) \rangle \longrightarrow \text{Br}(L).$$

Proof. Write $f_{\bullet} := \text{Ch}(f)$. A run over a chain e is a run of e 's shape (Lemma 4.2), and $f_{\bullet}e$ has that same shape with the classifying map forced on both sides, so $\mathcal{R}(e) = \mathcal{R}(f_{\bullet}e)$ as ordered sets — and everything in §3, Lemma 4.4 and Corollary 4.9 is defined from the wedge map alone. Hence $f_{\bullet}$ preserves $\perp$, $\top$, cells, steps, covers and γ , and it carries the sweep of the standard order on $\text{Sq}(e)$ to the sweep of the standard order on $\text{Sq}(f_{\bullet}e) = \text{Sq}(e)$, so it preserves src and tgt letter by letter: $\mathcal{P}(f)$ is a morphism of polygraphs, functorially in f because Ch is a functor.

Both composites send X to $[f_{\bullet}X]$. On a 1-cell α , $\text{Br}(f)$ is the descent of $Q_L \circ f_{\bullet}^{\text{op}}$, so it reads a cospan upstairs: $\text{Br}(f)\text{frac}(m, g) = \text{frac}(f_{\bullet}m, f_{\bullet}g)$. Here $f_{\bullet}\alpha_{\perp}$ is an order-preserving refinement $f_{\bullet}X \rightarrow f_{\bullet}\alpha$, hence equals $(f_{\bullet}\alpha)_{\perp}$ by Proposition 3.9, and $f_{\bullet}\alpha_{\top} = (f_{\bullet}\alpha)_{\top}$ likewise. So both composites send α to $\text{frac}((f_{\bullet}\alpha)_{\perp}, (f_{\bullet}\alpha)_{\top})$, and Proposition 2.2 finishes. The square commutes strictly — more than a presentation is entitled to — because an order-preserving refinement is pinned by its ends, so transporting a cell's legs involves no choice. $\square$

7 Theorem C: at the point the polygraph is Artin's

Since Z is terminal, every serial wedge has one bipointed map to it and every wedge map commutes over it, so $\text{Ch}(Z) \cong \square^{\vee}$: an object is a shape, a refinement a bare map of wedges. A run of Z is thus its event count; write Z_N for the run on N events. Both $\underline{e}$ and $\bar{e}$ are runs on $|e|$ events, so every cell of Z is a loop at $Z_{|e|}$.

Definition 7.1. For a shape a with $|a| = N$ put $\text{cut}(a) := \{k \in [N-1] : \text{the events of strands } k, k+1 \text{ share a bead}\}$, and $\text{cut}(d) := \text{cut}(\text{shape of } d)$. For $i \neq k$ in $[N-1]$ let $\text{cox}(i, k) := 3$ if $|i - k| = 1$ and 2 otherwise, the Coxeter exponent of S_N .

Lemma 7.2 ([Sta12, §1.2]). $a \mapsto \text{cut}(a)$ is a bijection from the shapes on N events onto the subsets of $[N-1]$, with $|\text{cut}(a)| = \deg a$. Hence $\text{Cell}_n(Z_N, Z_N) \cong \{C \subseteq [N-1] : |C| = n\}$: the 1-cells at

Z_N are indexed by $[N - 1]$ — write σ_k for the one with $\text{cut} = \{k\}$ — and the 2-cells by the pairs $i < k$. Moreover $|\text{Sq}(e)| = \text{cox}(i, k)$ for a degree-two e with $\text{cut}(e) = \{i < k\}$.

Proof. A shape on N events is a composition of N and $\text{cut}(a)$ is the complement of its bead boundaries in $[N - 1]$, so the first claim is the classical correspondence and $|\text{cut}(a)| = N - \#\text{beads} = \deg a$. For the last, e has a bead of dimension 3 exactly when three consecutive strands share one, that is when $k = i + 1$; then $|\text{Sq}(e)| = 3$ and otherwise 2 (Proposition 4.11), matching cox . $\square$

Lemma 7.3. For $s \in \text{St}(x)$ put $\text{pos}_x(s) := \min(\text{step}_x(\text{lo } s), \text{step}_x(\text{hi } s))$, so that x passes s between times $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$ (Lemma 4.4). For any chain d : (i) $\text{pos}_x : \text{St}(x) \rightarrow \text{cut}(d)$ is a bijection, for every $x \in \mathcal{R}(d)$; (ii) $\text{pos}_{x^s}(s) = \text{pos}_x(s)$; (iii) $\text{pos}_{\perp_d}(s) = \text{str}(\text{lo } s)$. If d is a chain of Z then moreover $\gamma(x \leq y) = \sigma_{\text{pos}_x(s)}$ for a cover with step s .

Proof. (i) The fibres of blk on the firing order are consecutive intervals of lengths $a_1, \dots, a_r$ (Lemma 3.3), which do not depend on x ; so the times t with t and $t + 1$ in one bead are $\text{cut}(d)$ for every x . A step is a cover inside a bead (Lemma 4.4), so pos_x lands there and is injective, distinct covers of a chain having distinct lower ends; and both sides have $\deg d$ elements (Lemma 4.4, Lemma 7.2). (ii) Flipping exchanges the times of the two events and changes no other. (iii) $\perp_d$ fires in strand order. For the last claim, $\gamma(x \leq y) = m_{\{s\}}$, whose beads are the blocks of the firing order cut at every cover but s (Proposition 4.6): one bead of dimension 2 at times $\text{pos}_x(s), \text{pos}_x(s) + 1$ and the rest edges, so its cut set is $\{\text{pos}_x(s)\}$. $\square$

Proposition 7.4. Let e be a chain of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$, and let $\perp_e = x_0 \leq \dots \leq x_{|\text{Sq}(e)|} = \top_e$ be the sweep of the standard order, with s_t the $(t + 1)$ -st square. Then $\text{pos}_{x_t}(s_t)$ is i for t even and k for t odd, so $\text{src}(e) = \sigma_i \sigma_k \sigma_i \dots$ and $\text{tgt}(e) = \sigma_k \sigma_i \sigma_k \dots$, both of length $\text{cox}(i, k)$.

Proof. By Lemma 7.3(i) each x_t has exactly two steps, passed at i and at k . For $t = 0$: $\text{pos}_{\perp_e}(s) = \text{str}(\text{lo } s)$ and the standard order sorts by $\text{str}(\text{lo } -)$ first, so if e is one bead of dimension 3 on strands $i, i + 1, i + 2$ its squares in order are $(i, i + 1), (i, i + 2), (i + 1, i + 2)$, with lower strands $i, i, i + 1 = k$; and if e is two beads of dimension 2 on $\{i, i + 1\}$ and $\{k, k + 1\}$ they are $(i, i + 1), (k, k + 1)$, with lower strands i, k . Either way the least square is passed at i and the greatest at k . For the step: s_t and s_{t+1} are both steps of x_{t+1} — the first because a flip of a step is a step, the second by Lemma 4.10 — and $\text{pos}_{x_{t+1}}(s_t) = \text{pos}_{x_t}(s_t)$ by Lemma 7.3(ii), so $\text{pos}_{x_{t+1}}(s_{t+1})$ is whichever of i, k that is not. The words follow by Lemma 7.3, and their length is $|\text{Sq}(e)| = \text{cox}(i, k)$ (Lemma 7.2). The dual order reverses the standard one, so its least square is the standard order's greatest, passed at k , and the same induction applies. $\square$

Definition 7.5. Artin's polygraph $\mathcal{P}_{\text{Artin}}$ has one 0-cell per $N \in \mathbb{N}$; at N , the loops $\sigma_1, \dots, \sigma_{N-1}$; and for each pair $i < k$ in $[N - 1]$ one 2-cell with the words $\sigma_i \sigma_k \sigma_i \dots$ and $\sigma_k \sigma_i \sigma_k \dots$ of length $\text{cox}(i, k)$, that is $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ when $k = i + 1$ and $\sigma_i \sigma_k = \sigma_k \sigma_i$ otherwise. These are Artin's relations [Art47], so the words at N modulo the 2-cells are the positive braid monoid B_N^+ ; since path composition reverses concatenation, $\text{End}(N) = (B_N^+)^{\text{op}}$ and $\langle \mathcal{P}_{\text{Artin}} \rangle = \mathbf{Braid}^{+, \text{op}}$.

Theorem 7.6 (Theorem C). $\mathcal{P}(Z) \cong \mathcal{P}_{\text{Artin}}$, cell for cell and word for word.

Proof. $\text{Run}(Z) \cong \mathbb{N}$, and every cell of Z is a loop, so both polygraphs are disjoint unions over N of loop polygraphs and it suffices to compare at each N . There $\text{Cell}_1 \cong [N - 1]$ and $\text{Cell}_2 \cong \{i < k\}$

by Lemma 7.2, matching Artin’s generators and relations, and by Proposition 7.4 the two words of the 2-cell at $\{i < k\}$ are Artin’s two words for that pair. A morphism of polygraphs bijective in every dimension is an isomorphism (Proposition 2.2). $\square$

Corollary 7.7. $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$, so $\text{Br}(Z)$ is the positive braid category up to word reversal. Moreover the terminal map gives $\text{Br}(!_K) : \text{Br}(K) \rightarrow \text{Br}(Z)$, a braid-valued invariant computed cell by cell: a degree-one chain of K goes to σ_k , where k is the position at which its run passes its one step, and a degree-two chain goes to the Artin relation of its two positions. This is the promised sharpening of [PZ21]: the braid is read off the chains of K directly, with no nerve and no realization. Note that no braid theory was used to get it — in particular Matsumoto’s theorem, which identifies B_N^+ with the Garside germ, is not needed.

Proof. Theorem 5.5 at $K = Z$, Theorem 7.6, and Theorem 6.1 at $!_K$. $\square$

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